Practical on molecular dynamics and binding energy calculations

Practical date: 22 January 2025
Report due: 5 February 2025
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Learning objectives

- Understand the steps involved in setting up a molecular dynamics (MD) simulation of protein-ligand in a box of water
- Develop an idea about how to analyze molecular dynamics simulations
- Understand binding energy calculations and the different components that effect these estimates

Guidelines for report

Please try to address the following questions in your report:

- **Simulation set up**: If you were to run a molecular dynamics simulation to analyze the effect of solvent in protein-drug binding interactions, what are the different entities that you have to fill the box with? Describe the steps involved in setting up a box for studying protein-drug binding. (1 point)
- **Binding energies**: Obtain an estimate of the interaction and solvation free energies of the protein and ligand for your trajectory. 0.5 point)
- **Trajectory analysis**: For the three additional trajectories provided, calculate the following quantities
  - Root mean square deviation of protein structure during trajectory compared to the initial frame and the distribution of RMSD
  - 2D RMSD of the trajectory (frame vs frame)
  - Root mean squared fluctuations of the protein residues during the trajectory
  - Distance of ligand from the binding pocket (or catalytic site residues)
  - Distance of ligand from specific residues of interest
  - Radius of gyration w.r.t time and its distribution
  - Ligand interaction plots for the interactions of protein residues and ligand
  - Principal Component Analysis with 2 components for the trajectory
  - Pearson Cross-correlation of the protein residues

  Using the calculated plots, comment on what is happening in the three trajectories. If the trajectories were carried out at temperatures T_1, T_2, T_3 respectively, how would you order these temperatures? Explain your analysis using all or a subset of the plots. (4.5 points)

NOTE Grades for this practical and report will be determined by the total points earned. The report covering the specified topics must not exceed 8 pages (including figures). No extra points will be awarded for unnecessarily long reports, and exceeding the page limit may result in a deduction of points.